Topic: Exponential type

Question: Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = \sin t$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t \to \infty} \frac{f(t)}{e^{\alpha t}} = 0$$

Answer choices:

A $\alpha > -1$

B $\alpha > 1$

C $\alpha > e^{\pi}$

D $\alpha > 0$

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Solution: D

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{\sin t}{e^{\alpha t}}$$

The value of $\sin t$ is never greater than 1, but $e^{\alpha t}$ grows exponentially for any $\alpha > 0$. So the denominator will grow infinitely large, while the numerator remains small, so the limit will go to 0, and the Laplace transform $\mathscr{L}(f(t))$ is defined. Therefore $f(t) = \sin t$ is of exponential type for any $a > 0$.

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Topic: Exponential type

Question: Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = t^n$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t \to \infty} \frac{f(t)}{e^{\alpha t}} = 0$$

Answer choices:

A $\alpha > 1$

B $\alpha > 0$

C $\alpha > \ln n$

D $\alpha > -1$

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Solution: B

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{t^n}{e^{\alpha t}}$$

For the numerator, given any real number x, we know that

$$e^{\ln x} = x \quad \text{and} \quad \ln(e^x) = x$$

so we can write the limit as

$$\lim_{t\to\infty} \frac{t^n}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{e^{\ln(t^n)}}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{e^{n\ln t}}{e^{\alpha t}} \to \lim_{t\to\infty} e^{n\ln t - \alpha t}$$

We know that the only way for this limit to approach 0 is if the exponent tends to $-\infty$, in other words, if $n\ln t - \alpha t \to -\infty$ as $t \to \infty$.

This is always true, since logarithmic growth is always slower than linear growth. Therefore the function $f(t) = t^n$, for a fixed positive integer n, is of exponential type for any $\alpha > 0$.

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Topic: Exponential type

Question: Determine the value of α in $e^{\alpha t}$ such that the function $f(t) = b^t$ is of exponential type. In other words, find the value of α that makes the following equation true.

$$\lim_{t \to \infty} \frac{f(t)}{e^{\alpha t}} = 0$$

Answer choices:

A $\alpha > 1$

B $\alpha > 0$

C $\alpha > \ln b$

D $\alpha > -1$

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Solution: C

Substituting $f(t)$ into the limit gives

$$\lim_{t\to\infty} \frac{f(t)}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{b^t}{e^{\alpha t}}$$

With this we can write

$$\lim_{t\to\infty} \frac{b^t}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{e^{\ln(b^t)}}{e^{\alpha t}} \to \lim_{t\to\infty} \frac{e^{t\ln b}}{e^{\alpha t}} \to \lim_{t\to\infty} e^{t(\ln b-\alpha)}$$

We know that the only way for this limit to approach 0 is if the coefficient of t in the exponent is negative. In other words, if $\ln b - \alpha < 0$. Therefore the function $f(t) = b^t$, for a fixed $b > 0$, is of exponential type for any $\alpha > \ln b$.

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